

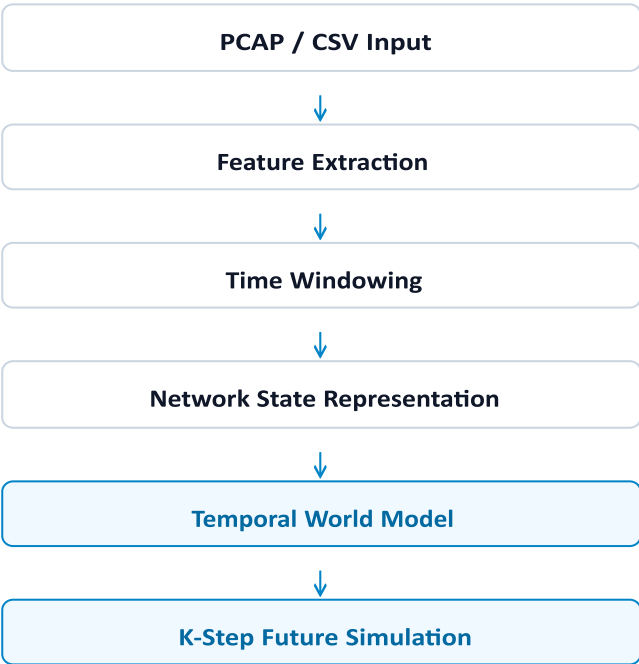
CyberWM

AI World Model for Predictive Cyber Defence

1 PROBLEM STATEMENT

Traditional intrusion detection systems commonly classify individual traffic flows as benign or malicious. CyberWM instead models network activity as a sequence of evolving states and predicts future attack progression.

2 SYSTEM ARCHITECTURE



PCAP / CSV

Ingests raw packet captures or structured flow logs from network interfaces.

Feature Extraction

Parses headers, payload attributes, and statistical flow metrics into structured vectors.

Time Windowing

Aggregates sequential features into uniform temporal buckets to track dynamics.

Network State

Constructs compact latent vectors representing current system status and topology.

Temporal World Model

Learns environment transitions to forecast state evolution under latent adversarial dynamics.

K-Step Future Simulation

Projects trajectories K steps forward to anticipate multi-stage threat progression.

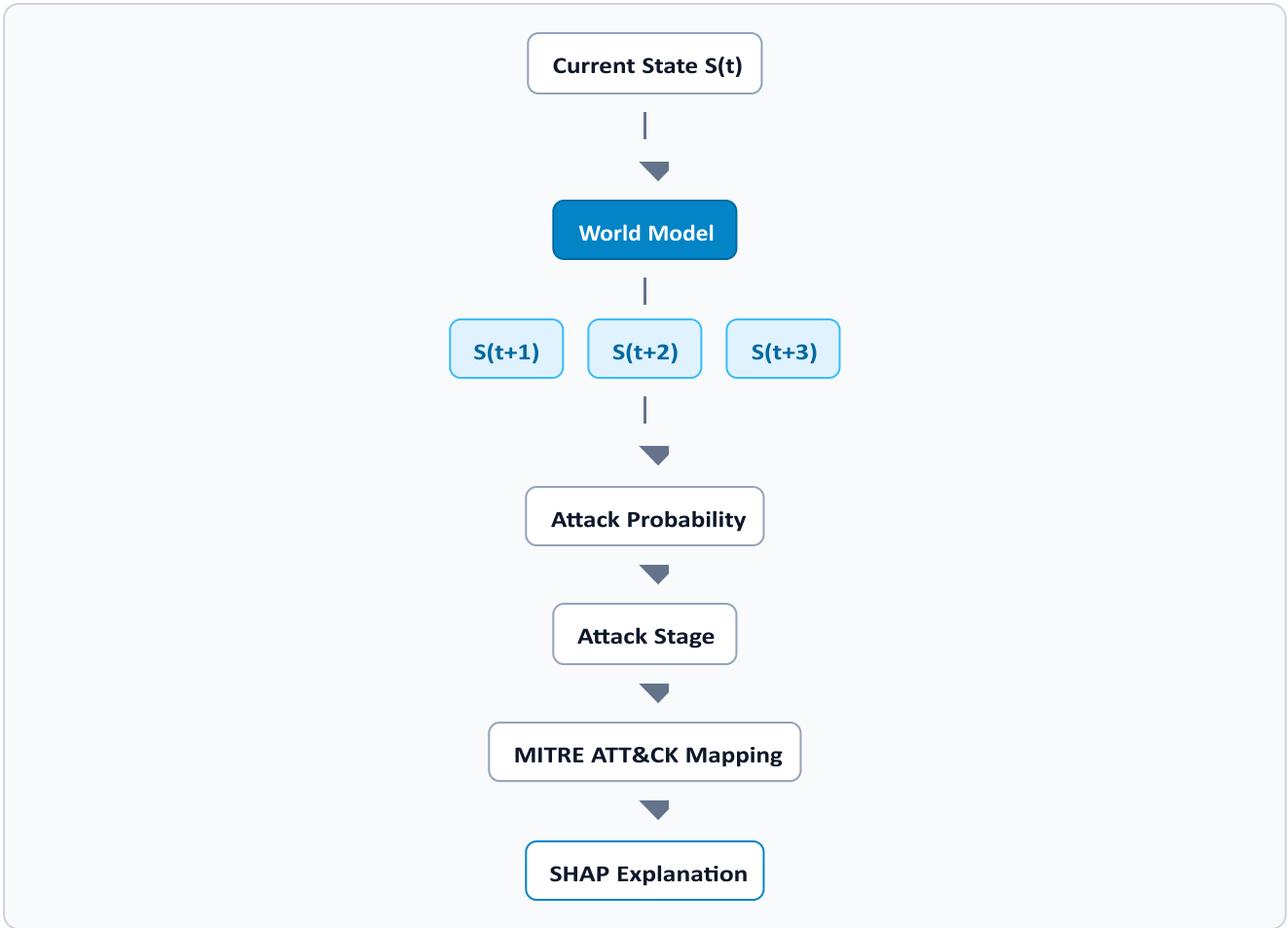
3 TECHNOLOGY STACK

PYTHON

PyTorch

Pandas

4 AI & PREDICTION PIPELINE



5 MODEL OUTPUTS

1. Current Attack Probability	2. Future Attack Probability
3. Predicted Attack Stage	4. MITRE ATT&CK Mapping
5. Top Contributing Features	6. Risk Timeline

6 EXPERIMENTAL EVALUATION

Comparative performance evaluation on benchmark sequential network traffic datasets.

Model Architecture	Precision	Recall	F1 Score	False Positive Rate (FPR)
Logistic Regression (Baseline)	0.812	0.764	0.787	0.084
CyberWM (Temporal World Model)	0.954	0.938	0.946	0.012